



1 (a) Complete the sentences about number systems.

Use the items from the list.

Some of the items in the list will **not** be used. You should only use an item once.

A	B	C	D	E	F	G
H	W	X	Y	Z	0	1
2	4	8	10	16	127	128
255	256					

The binary number system is base The smallest denary number that can be represented as an 8-bit binary number is The largest denary number that can be represented as an 8-bit binary number is

The hexadecimal number system is base Each hexadecimal digit is equivalent to bits. The numbers 1 to 9 are used and the number 10 is represented by The hexadecimal number system continues up to the number 15, which is represented by

[7]

(b) Two 8-bit binary numbers are given.

Add the two 8-bit binary numbers using binary addition.

Give your answer in binary. Show all your working.

$$\begin{array}{r} 10011011 \\ + 00010011 \\ \hline \end{array}$$

[3]

